

Alex Lee

Waterloo, Canada | a55lee@uwaterloo.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SKILLS

Languages: Python, Java, C/C++, C#, R, SQL(MySQL, MongoDB), Bash, Typescript, Javascript

Data/ML: Pandas, Matplotlib, SciKit, TensorFlow, NumPy, OpenCV

Tools: React, Express, Next.js, Spring, Selenium, Bootstrap, SASS, Excel, PowerBI, Git, AWS

EXPERIENCE

Machine Learning Engineer

Jan. 2026 – Aug. 2026

City of Edmonton - Future Cities Institute

Edmonton, AB

- Architected and deployed **CityVision**, an open-source real-time traffic control software, achieving **65% cost savings** in city infrastructure operations through automated vehicle detection.
- Optimized a **YOLOv11** vehicle detection model with an automated data pipeline on **GCP** for continuous training, improving model inference efficiency and lifecycle management.
- Engineered a real-time monitoring dashboard using **Shiny R** to provide stakeholders with interactive traffic analytics and live insights for infrastructure decision-making.
- Containerized the entire application stack using **Docker** to ensure a scalable environment and streamlined deployment via **CI/CD pipelines** (GitHub Actions/GCP).
- Collaborated in an **Agile** environment, regularly presenting technical progress and model performance metrics to stakeholders to facilitate data-driven urban policy improvements.

Fullstack Developer

May. 2025 - Aug. 2025

Association of Korean Canadian Scientists and Engineers

Waterloo, ON

- Led the **team of 6** to develop the official AKCSE website using **Express** and **MongoDB**, improving accessibility to career and academic help resources for students.
- Optimized query efficiency and database scalability by **300%** through implementation of **dynamic CRUD**, while implementing **authentication** and access control features to strengthen data security.

PROJECTS

Computer Vision Football Analytics | [GitHub](#) | [Link](#)

- Trained a **YOLOv11** model with **10,000+** images, achieving **0.877** mAP50 with 35.7 ms per-frame latency.
- Deployed services on **AWS EC2** with **SQS**, and built a **Next.js** web app for video upload and real-time analysis.

Credit Card Fraud Detection | [GitHub](#)

- Developed credit card fraud classification model using **sklearn** and **TensorFlow** with **96%** prediction accuracy.
- Processed and cleaned **200,000+** transaction records, focusing on handling imbalanced financial datasets.

Customer Churn Prediction Model | *Python, SciKit, Pandas, Streamlit* | [GitHub](#)

- Built a **machine learning model** to predict customer churn, using **900+** telecom industry customers' datasets.
- Utilized the model with interactive **Streamlit** web app to improve accessibility and gain actionable insights.
- Identified **central factors**(tenure, monthly charges), helping companies plan their products more effectively.

FBREF Football Data Analysis | [GitHub](#)

- Collected **100,000+** football match data using **Selenium** and organized the data into a **Pandas DataFrame**.
- Identified **progress passing distance** as key predictor of match results, suggesting focused strategies to teams.

Academic Management System | *Spring, Javascript, HTML/CSS, MySQL, jQuery, Bootstrap* | [GitHub](#)

- Built a full-stack academic management web app of **20+** pages with **50+** users using **Spring** and **MySQL**.
- Distributed project on **AWS EC2** with **Apache Tomcat**, with optimized database schema and RESTful APIs.
- Implemented **10+** essential features, **CRUD** operations of course postings and assignments, self implemented real-time chat, custom course search filters, and dynamic operations and views with the database.
- Reduced query response time by **30%** using nested SELECT statements to handle grouped ID increments.
- Ensured reliability with **200+** test cases and resolved issues such as multiple PK, time zone mismatches.

Personal Portfolio Website | *Typescript, React, SASS* | [GitHub](#) | [Link](#)

- Developed a personal portfolio website using **Typescript** and **React** to showcase professional skills and projects.
- Designed a modern portfolio website using **SASS**, including responsive layouts, for better user experience.
- Enhanced personal brand visibility and resulted in **150%** increased networking and interview opportunities.

Biquadris Game | *C++*, *X-Window* | [GitHub](#)

- Led a team of three to develop a non-real time Tetris as a CS246 final project, using **C++** with OOP design.
- Built a visually responsive game board with real-time score tracking and piece previews using **X-Window**.
- Conducted over **500** test cases to ensure stability and reliability to reduce runtime errors during play sessions.

EDUCATION

University of Waterloo

Sep. 2023 - Apr. 2028

Bachelor of Mathematics, Major in Data Science

Waterloo, ON

Minors: Combinatorics and Optimization, Management

Relevant Coursework: Data Structures, Algorithms, OOP, Statistics, Linear Models